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Effects of Label Noise on Performance of Remote Sensing and Deep Learning-Based Water Body Segmentation Models

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ABSTRACT

Large-scale management of surface water resources in urban areas can be difficult, especially if the region is subject to monsoonal waterlogging. Deep learning-based methods for computer vision tasks, such as image segmentation, can effectively be applied to remote sensing data for generating water body maps of large cities, aiding managerial entities, urban planners and policymakers. The robustness of these models to erroneous pixel level training class labels has not been studied in water body segmentation context. Label noise is commonly experienced in classification tasks and may hinder the performance. We collected and densely labeled Sentinel-2 images over Dhaka, one of the most densely populated and flood prone megacities in the world. We synthetically injected four types of label noise viz., i) Gaussian noise, ii) translation, ii) rotation, and iv) mirroring. Our primary objective is to observe and quantitatively analyze the effects of label noise on remote sensing data-driven deep learning models for water body segmentation. Our results show that salt and pepper noise (injected artificially using Gaussian noise) of only 50% can cause a massive 48.55% drop in Intersection-over-Union score. The consequences of learning from training data with different magnitudes and settings of label noise have been explored.

KEYWORDS

Deep learning; image segmentation; label noise; remote sensing; water body extraction

Introduction

Waterbodies play a vital role in urban ecosystems, for example, ecosystem services such as precipitation/rainfall, surface energy, and climatic conditions of urban areas (Carroll et al. 2016). Therefore, management and maintenance of surface water is important. In recent decades, developing countries like Bangladesh, India, Vietnam etc. are under rapid urbanization, but most of this urbanization is not well planned (Goldblatt et al. 2016).

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The capital of Bangladesh, Dhaka, is one of the most rapidly growing cities in the world with over 10 million inhabitants (Faisal and Parveen 2004). Aggressive urbanization has caused large-scale land cover changes (Afroz 2016). Due to its geographical conditions, Dhaka faces heavy rainfall in the monsoon months. This rainwater is generally drained by the stream and canals (Rahman and Hossain 2008), which then flow downstream to nearby rivers, such as the Buriganga, Balu, Turag etc. rivers. Due to rapid urbanization, these low-lying water bodies are quickly filled up (Maksud Kamal and Midorikawa 2004). As a result, disruption of water flow creates severe waterlogging under even light rainfall. On the other hand, aquatic vegetation play a significant role in controlling pollution (Rahman and Hossain 2008). For these reasons, regular mapping and monitoring of water bodies are necessary for cities in this region.

Remote sensing technology has enabled large-scale and high-resolution land-cover and land-use mapping (Weng 2002). Previously, high-resolution remote sensing images were not readily available, but now, sensors onboard various satellites such as Sentinel-2, Landsat-8, Gaofeng-2 etc. can provide very detailed images which can allow the generation of very accurate maps (Zeng et al. 2015). As a result, applications of remote sensing data have boomed. For example, mapping reservoirs, land use, land cover, agriculture (Jawak, Kulkarni, and Luis 2015; Yang et al. 2017). Waterbody extraction from Sentinel-2 imagery is well studied in the literature (Y. Du et al. 2016; Yang et al. 2017). Composite band indices been developed for water body extraction from remote sensing images (Xu 2006). However, these indices were found to be erroneous for predicting waterbodies. In order to resolve these issues, the Convolutional Neural Network (CNN) based models have emerged as the state-of-the-art (Chen et al. 2018; Isikdogan, Bovik, and Passalacqua 2017).

Label noise can be issue in image segmentation tasks since dense annotations are labor intensive to label and are prone to having imperfections. Label noise can be introduced during the manual labeling or through artificially injected noise, and during inference when labels are generated. The concept of label noise arises from distinction of ground truth labels (which are idealistic and perfectly labeled noise-free clean labels and usually unobserved and unavailable) and observed labels (which are labels that are accessible but have erroneous labels), in practice these labels can capture important embedding. Label noise occurs whenever the ground truth label and observed label are not equal (Frénay and Verleysen 2014; Li, Huang, and Chang 2020).

Classification performance of models trained on data corrupted by noisy labels is studied in most works. Label noise can naturally be introduced in any process involving annotation by humans. Label noise robust land cover

mapping models incorporating deep learning can leverage remote sensing data (Li, Huang, and Chang 2020). The accuracy has been reported to be low even when robust classification models are trained on noisy data (Li, Huang, and Chang 2020). To the best of our knowledge, the effects of label noise on performance of deep learning-based water body segmentation models have not been studied and/or reported yet.

In this study, we observe and analyze the consequences of different types of simulated label noise on deep learning-based urban water body image segmentation tasks, utilizing Sentinel-2 images. We use the U-Net (Ronneberger, Fischer, and Brox 2015), an autoencoder architecture with skip connections, as our segmentation model. We create different training datasets with simulated label noise and gauge its effects on classification performance of the U-Net. Our major contributions are as follows:

1. We outline methods for simulating salt and pepper errors using Gaussian noise and registration errors using image translation and also delve into less commonly observed label noise types such as mirroring.
2. We have quantified the performance of deep learning-based water body segmentation models using several performance metrics tailored for reliable inferences on imbalanced datasets.
3. We have labeled, curated and open sourced our dataset of Sentinel-2 images, corresponding clean labels and labels with simulated noisy labels (https://github.com/monsurhillas007/LabelNoisePerformanceonWaterBodySegmentation).

Background and Related Work

Remote Sensing and Water Body Mapping

Remote sensing refers to collection of information about objects in the physical environment without any physical contact; the collection (sensing) occurs in noncontact (remote) fashion. Properties of radiation (such as wavelength and intensity/magnitude) from physical objects can be analyzed for classifying land use and land cover classes.

Li, Huang, and Chang (2020) proposed various methods which use remote sensing imagery for detecting waterbodies, viz., single based threshold, multi-band threshold, supervised classification and unsupervised classification approaches. Isikdogan, Bovik, and Passalacqua (2017) proposed the data-driven and deep learning based DeepWaterMap model for mapping surface water, which performed better than the previously implemented normalized difference water index (NDWI) based method. Chen et al. (2018) used adaptive simple linear iterative clustering (A-SLIC) algorithm, by minimizing the reliance on segmentation of remote sensing images in

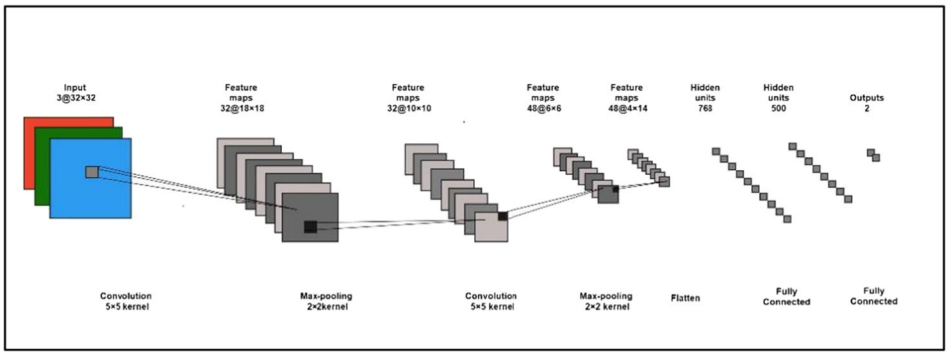


Figure 1. Architecture of the CNN used in Cheikh Sidiya and Li (2020).

initial state. A modified convolution neural network SAPCNN has been proposed by Chen et al. (2018) to be used for water extraction from satellite images. It was observed that, normalized difference index threshold methods and support vector machine classification methods were poor at differentiating between shadow and water pixels, whereas the SAPCNN model was able to make clear distinctions.

Convolutional Neural Networks and Fully Convolutional Networks

Convolutional Neural Networks (CNNs) employ architecture level structural parameter tuning, which allows the learning of complex and nonlinear relationships between images and labels. This flexibility allows better fits to many different forms and types of data (O'Shea and Nash 2015). The convolutional layers extract higher order features from inputs (Chen et al. 2018). Figure 1 shows a generic CNN architecture with relevance to feature maps and convolutions.

However, the general CNN is engineered for image classification tasks; usually, a single scalar like class label is assigned to an image. For image segmentation tasks, the CNN has been architecturally modified for the purpose of dense prediction. Fully convolutional networks (FCNs) are autoencoder like architectures with skip connections and traditionally have generic CNNs as backbone of the FCN. The most popular FCN architecture is the U-Net, and has been applied for image segmentation tasks in a variety of fields. As a result, there are also many variants ('flavors') of U-Nets which have been developed and applied. Feng et al. (2019) developed deep learning based water body extraction using remote sensing data and a modified U-Net. The APP-UNet model (P. Zhang et al. 2018) (obviously based and named after the U-Net), outperformed other methods (P. Zhang et al. 2018).

Application of U-Net

The original U-Net (Ronneberger, Fischer, and Brox 2015) was published in 2015, since then it has widely been adopted in the biomedical sector. Falk et al. (2019) came up with a web plugin that enables any medical personnel to use the trained U-Net model for cell counting and detection. A novel variant of U-Net has been used for cardiac image segmentation by (Zheng et al. 2018) where the segmented a slice (from top) of a cardiac image from an already existing adjacent slice. Many other segmentation use cases in medical sector has been aided by U-Net and its variants. Rectal tumor from MR images (J. Wang et al. 2018), liver vassal segmentation from CT images (Huang et al. 2018), cartilage and meniscus segmentation (G. Du et al. 2020), lung segmentation from chest X-ray (Gaál, Maga, and Lukács 2020) and many other (Siddique et al. 2020) medical image segmentation use cases has been successfully encountered using U-Net and its variants. Being encouraged by the success of U-Net in medical image segmentation, the research community came up with many other applications of this architecture in different sectors. It has been showed the usage of U-Net to segment sea-land (Chu et al. 2019) while replacing the contracting path of the original U-Net with a ResNet architecture. The study showed how U-Net can be used to separate a singing voice from a musical track (Jansson et al. 2017). Segmentation of root from soil (Smith et al. 2020), road extraction (Zhang, Liu, and Wang 2018) and tomato leaf segmentation (Ngugi, Abdelwahab, and Abo-Zahhad 2020) are some examples of non-medical cases where U-Net has been applied for segmentation task. From the discussion above, it can be said that, many different variants and instance specific modifications, U-Net is being widely used for the past few years. Our study is mainly directed toward finding the effect of different levels of varying noise on U-Net as U-Net is the core basic and all the other modification are instance specific modifications of the original architecture.

Label Noise

Label noise can be categorized into three major types such viz., i) *Noisy Completely at Random* (NCAR) labels, ii) *Noisy at Random* (NAR) labels and iii) *Noisy not at Random* (NNAR) labels (Frénay and Verleysen 2014). In the NCAR scenario, the observed label is independent of the true class label and the features. In the NAR scenario, the observed label is dependent on the true class label but still independent of features. In the NNAR scenario, the observed label is dependent on both the true class label and the features (Frénay and Verleysen 2014; Frank et al. 2017). It is evident, the type of label noise models is characterized and determined by the randomness/stochastic nature of the label noise process. Nonrandom behavior is encoded by dependencies on true class label and/or features.

Related Work on Water-Body Detection and Label Noise

In recent years, water bodies in urban areas are decreasing in size and numbers, making them harder to identify and track down using remote sensing approaches. Nevertheless, studies continue to focus on remote sensing-based water body detection, as it is of great managerial importance. Many works have been done related to water body detection. Zhang, Liu, and Wang (2018) compared the DenseNet model with state-of-the-art models like VGG, SegNet, DeepLabV3 and ResNet. G. Wang, Wu, et al. (2020) focused on the Poyang Lake, located in Jiangxi province of China, using CNN and remotely sensed data. Mapping land cover change in the presence of label noise has been studied for Shenzen (in China) (Li, Huang, and Chang 2020).

Materials and Methods

Data

Image Collection

Most of Dhaka city's water bodies, such as rivers and canals, are connected to neighbor districts like Gazipur and Narayanganj. We have included the suburban area neighboring with Dhaka (such as Savar and Keraniganj). The georeferenced Sentinel-2 images have been collected from Google Earth Engine. The study area boundary shapefile is extracted from Angel et al. (2012). We limited our search of Sentinel-2 images to those covering/intersecting the study area boundary shapefile and to images of winter of 2019. We then form a composite, which is clipped using the study area boundary shapefile. Dhaka receives regular precipitation throughout the monsoon and intermittently throughout other season except winter, and the presence of clouds may obscure the legibility of multi-spectral remote sensing data. For our work, we use the bands with the highest resolution, i.e., 10 m (B2, B3, and B4 bands).

Labeling and Preprocessing

Water bodies and non-water bodies of Dhaka city are categorized on the Google Earth Engine (GEE) platform (Gorelick et al. 2017). Waterbodies are photo-interpreted, identified and labeled using the GEE polygon drawing tool. More than 50% of water bodies are labeled manually (Figure 2). Afterwards the water body polygon labels are exported to local storage. The non-waterbody polygon labels are generated by clipping away the waterbody polygons from the study area boundary polygon. The two polygon shapefiles are merged and then rasterized at the same resolution as the input satellite image (10 m) and coordinate reference system (Figure 3). We tile the large satellite image raster to tiles of physical dimensions of

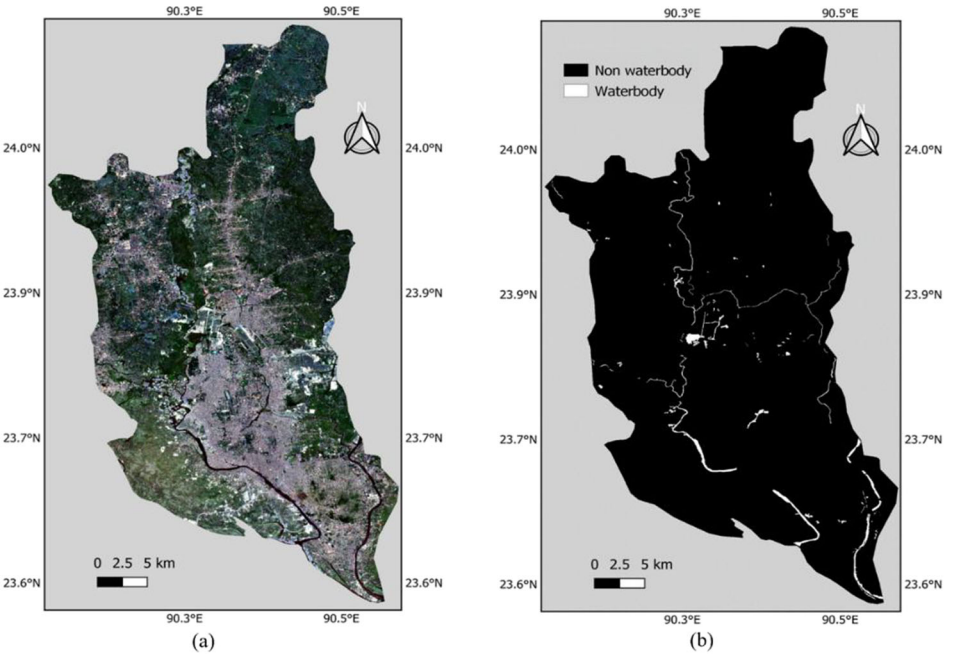


Figure 2. (a) Rasterized Image, (b) Vector merged image.

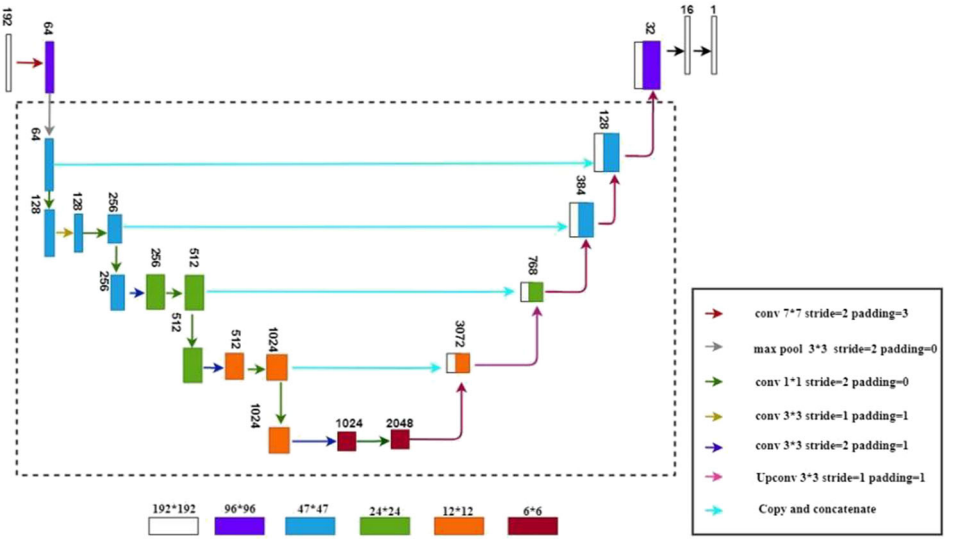


Figure 3. Network architecture of the U-Net segmentation model we have used.

1920 m × 1920 m and image dimensions of 192 × 192 pixels. We convert the pixel values of the 3 bands to 8-bit unsigned integers. Tiling at 192 × 192 pixels leads to 756 tiles, out of which only 110 have at least one or more waterbody pixels (and are thus selected for use in our work). The remaining are all tiles without any waterbody pixels.

Methods

Segmentation Model

The task of water body segmentation can be represented as a generalized image segmentation problem with binary pixel level labels (i.e. every pixel is either a waterbody or a non-waterbody). Currently, deep learning models provide state of the art performance for general computer vision tasks, as well as in the field of remote sensing. There are several options to choose from, such as the FCN. It is important to note that, these architectures can sometimes easily be transformed to exactly imitate other architectures using hyperparameter level tuning. The FCN can be thought of as a generalized U-Net and thus can be tuned to represent the U-Net as well.

Network Architecture

The autoencoder like architecture of the U-Net consists of an encoder (which is usually replaced by a CNN architecture also known as the 'backbone') and a decoder (which applies the deconvolution operation and interpolation methods to restore the data to initial height and width dimensions) with skip connections. We use the ResNet50 architecture, which has been specifically designed to solve vanishing gradient issues by introducing residual blocks. The soft label predictions represent probabilities and thus lie in the range 0-1. Therefore, we use the sigmoid activation function in the final layer. The sigmoid activation function takes any real value input (no matter how big or small) and squeezes the embedding data to a range between 0 and 1. The hard labels are obtained using a threshold value which determines the final class label of each pixel.

Loss Function

Loss functions have a significant role in the performance of a deep neural network. The optimizer used during training a neural network model updates all the parameters in such a way, that the loss is minimized. This is very essential part of training a deep neural network which eventually creates great impact on the learning and performance of a model. So, choosing the appropriate loss function is an important decision. We chose four loss functions namely BCE, Tversky, Focal and dice loss which was previously proven as noise robust in many studies (Jadon et al. 2020; Ghosh, Kumar, and Sastry 2017). We tried these loss functions in our context of study.

Binary Cross Entropy is a distribution based loss function and one of the most commonly used loss function in the classification and segmentation model (Ma, Liu, and Qian 2004). In Binary Cross Entropy the model has to predict much confidently about the class (Jadon et al. 2020).

$$CE\Big(p,\ y) = \begin{cases} -\log(p), ify = 1 \\ -\log(1-p),\ otherwise \end{cases} \tag{1}$$

Here, in this Equation (1) p, y denoted as output probability from the model and ground truth of the class respectively. This loss function will only provide a true positive class if the probability value for that class is greater than 0.5. This model is not enough noise robust when the data samples are class imbalanced.

With utmost attention, we evaluated our model with three new proven noise robust loss functions. First, we have tested with the dice loss in our model. Dice Loss itself is a noise robust loss function and one of the most frequently used loss functions along with U-Net (L. Wang, Wang, et al. 2020).

$$Dice\ \ Loss = \frac{2|X \cap Y|}{|\mathrm{X}| + |Y|} \tag{2}$$

Here (Equation (2)), X and Y represent the predicted output and ground truth value respectively. Dice loss can optimize the dice coefficient directly and it does not require any class-based re-weighting like Binary Cross Entropy (G. Wang, Liu, et al. 2020). Usually, Dice loss works well for the binary class segmentation model. We have also implemented two more noise robust loss (Tversky, Focal) to check how they perform against dice loss.

Tversky is a region based loss function where the main objective is to minimize the mismatch and maximize the common region between ground truth and the predicted segmentation (Salehi, Deniz Erdogmus, and Gholipour 2017). To achieve a better tradeoff between precision and recall Tversky is a better option than other loss functions.

$$Tversky\ \ Index = \frac{|\mathrm{XY}|}{|\mathrm{XY}| + \alpha|\mathrm{X\ Y}| + \beta|\mathrm{Y\ X}|} \tag{3}$$

Here (Equation (3)), X and Y are also denoted as output and ground truth respectively. Unlike dice loss, α, β is presented here as weight penalty for False Positive (FP) and False Negative (FN) respectively. This α, β values control the magnitude of FP and FN rate. A highly imbalanced dataset like a noisy labeled dataset, where FP and FN high rate is a common situation. So, Tversky loss makes better prediction by penalizing this high rate of FP and FN values. We have kept the value of α, β as 0.5.

Here, as a loss function, we have also used Focal loss. Focal loss is a distribution based loss like Binary Cross Entropy (Lin et al. 2020). But where Binary Cross Entropy shows vulnerability for class imbalance dataset, focal addresses these issues with adding extra weighted penalty. To derive the

focal loss, we can also write the equation (Equation (1)) as below,

$$Pt = \begin{cases} p, ify = 1 \\ 1-p, \ otherwise \end{cases} \tag{4}$$

This can also be called as the probability of class. Using this probability, the equation for focal loss can be denoted as,

$$Focal\ Loss(pt) = -\alpha_t(1-pt)^{\gamma}\log(pt) \tag{5}$$

Here, if γ range is greater than 0 and if $\gamma = 1$ Focal Loss behaves as Binary Cross Entropy. On the other hand, α is to handle the imbalance class issue. This weighted value considers the soft and hard sample which was missing in BCE and makes it more noise robust then other loss function. We have used $\alpha = 0.8$ and $\gamma = 2$.

Label Noise Injection

We considered two most commonly encountered sources of errors in remotely sensed data – i) salt and pepper noise and ii) registration errors; we also consider a more human-only source of errors, mirroring/flipping of label raster. We use pre-trained versions of commonly used image processing methods freely available as software packages with slight modifications. We also simulate registration errors using two different image processing methods – a) translation/shift and b) rotation. To summarize, we generate training datasets with the following label noise injection methods:

- Gaussian noise in label
- Translation/shift induced label noise
- Rotation induced label noise
- Mirroring/flipping induced label noise

Gaussian Noise in label: Addition of Gaussian noise in label follows the NCAR model. The only parameter required is the noise rate i.e. the proportion of examples/pixels which are noisy or to be noised. This proportion (expressed as %) or percentage determines the amount of noise in each label tile. We generate datasets with noise rates in the set {1, 5, 10, 15, 20, 25, 30, 40, 50, 60} (Figure 4). As our dataset is heavily imbalanced where waterbody pixels are much less frequent, the addition of Gaussian noise increases the amount of observed water body pixels. Owing to the binary classification setting, flipping the label is nontrivial i.e. 0 is flipped to 1 and 1 is flipped to 0.

Translation induced label noise: Translation refers to the shift/repositioning of the observed label with respect to the original label. We shift the labels

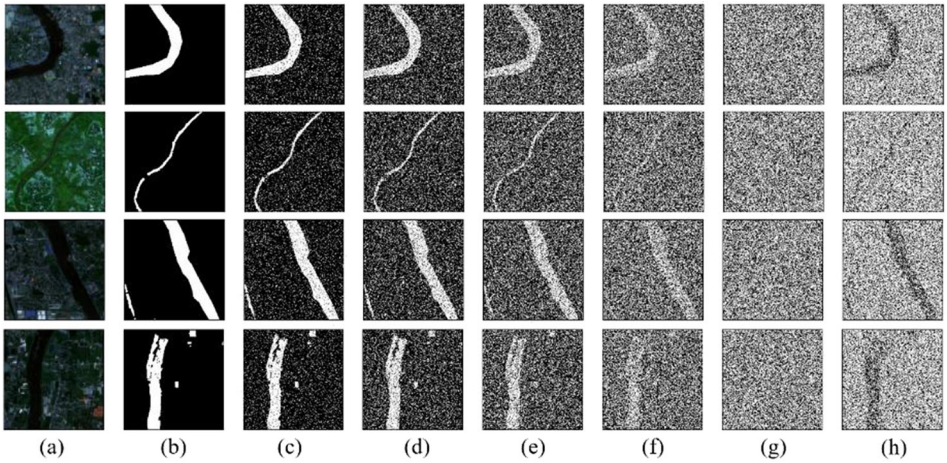


Figure 4. Some examples of label tiles with injected Gaussian noise (a) Image, (b) Ground truth label, (c) injected with 10% Gaussian noise, (d) injected with 20% Gaussian noise, (e) injected with 30% Gaussian Noise, (f) injected with 40% Gaussian noise, (g) injected with 50% Gaussian noise, and (h) injected with 60% Gaussian noise.

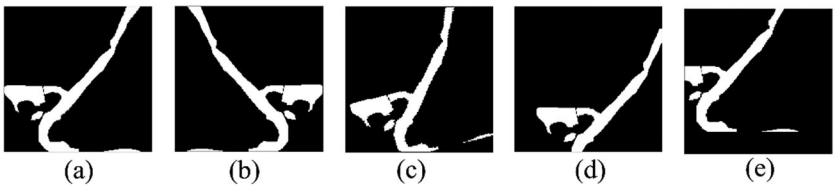


Figure 5. Label noise induced by translation (a) Ground truth, (b) Horizontal Flip of the label, (c) 15 degree right rotate of label, (d) 15% positive shift of label and (e) 15% negative shift of label.

along the positive and negative x-axis, i.e., horizontally, where the original label tile is situated at the origin. The tiles are shifted relative to the dimensions of the label tiles and are parameterized by percentages (shift amount as a percentage of width); we generate training data with labels shifted by values in the set $\{5, 10, 15, 20\}$. While shifting, we either do a bottom-right/positive shift (where label tiles are shifted in the direction of the positive x-axis and negative y-axis) or a top-left/negative shift (where label tiles are shifted in the direction of the negative x-axis and positive y-axis). We also set the probability of a label tile being flipped to 0.5 and 1.0. This probability parameter essentially governs the number of flipped labels in the dataset. Figure 5 shows some examples of label tiles with translation; it is apparent that this type of noise injection closely resembles registration errors.

Rotation induced label noise: Another way to express registration errors is through rotation of label tiles. This model has two parameters – i) the rotation amount (in degrees) and ii) the probability of a label tile being rotated. The label tiles are rotated by 5, 10 and 15 degrees. The probability values

selected are 0.5 and 1.0. Here, probability, $p = 1$ means 100% of the dataset consists of noisy label and $p = 0.5$ means 50% data consists with label noise Therefore, six different training datasets with rotated labels. Figure 5 demonstrates the image rotation for some waterbodies.

Mirroring induced label noise: Mirroring of label can also be performed by flipping the label tiles. We flip the label tiles along the x-axis, basically mirroring the label tile on the y-axis and causing a horizontal flip. This model is parameterized by the probability of a label tile being flipped. We generate datasets with 25%, 50% and 100% of label tiles horizontally mirrored.

Results and Discussion

Performance Evaluation Metrics

We calculate the number of true-positive (TP), true-negative (TN), false-positive (FP), and false-negative (FN) pixels; the independent test set had 20 image tiles (each 128×128 pixels). We have evaluated our performance based on five evaluation metrics. They are, Accuracy (Equation (6)), Intersection Over Union (Equation (7)), F1-score (Equation (8)), Precision (Equation (9)), and Recall (Equation (10)). Other than Accuracy, the other metrics are especially suited for evaluating imbalanced datasets.

$$Accuracy = \frac{TP + TN}{TP + TN + FN + FP} \tag{6}$$

$$IoU = \frac{TP}{TP + FN + FP} \tag{7}$$

$$F1 - score = \frac{2TP}{2TP + FP + FN} \tag{8}$$

$$Precision = \frac{TP}{TP + FP} \tag{9}$$

$$Recall = \frac{TP}{TP + FN} \tag{10}$$

Performance under Noise-Free Settings

We first evaluate our model on a noise free scenario for establishing the baseline performance of our model. The baseline performance in a noise free setting will be used as our threshold for comparing the performance of our model under different noise while varying the amount of noise presence. As the baseline setup, we chose dice as our loss function and Adam as our optimizer which is empirically found to be performing well on our study. As stated previously, we calculate the number of TP, FP, FN and TN

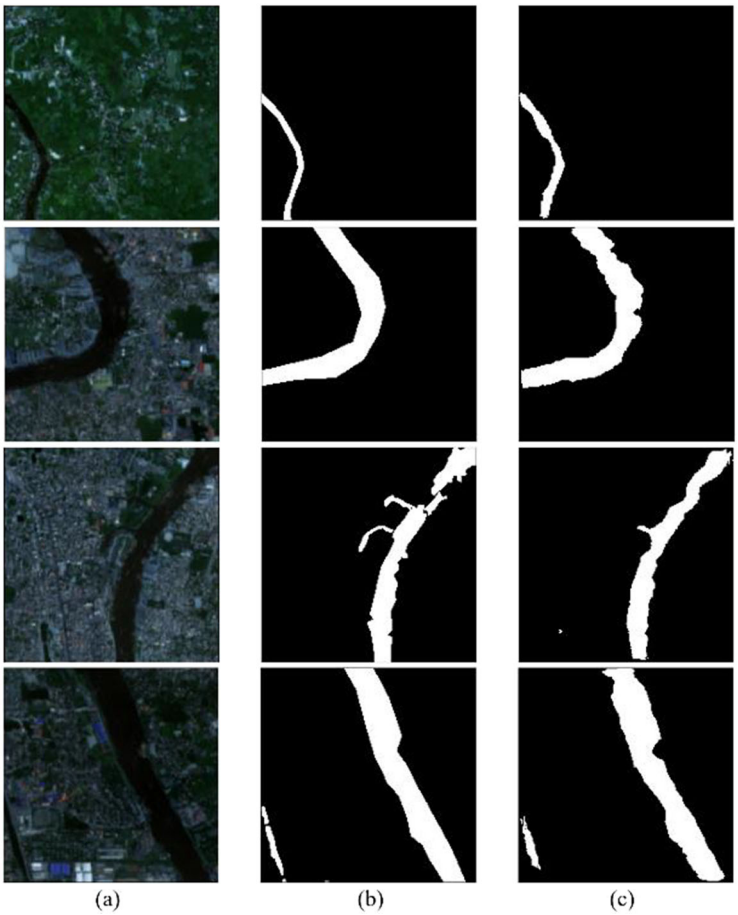


Figure 6. Some examples of predictions from model trained on data without any added label noise (a) Images, (b) Ground truth labels, (c) Predicted labels.

predicted pixels on the test set using equations stated in section "Performance Evaluation Metrics". The result of baseline model shows accuracy of 97.71%, and IoU 64.99%, F1-score 78.78%, Precision 80.23%, and Recall of 77.38%. Some qualitative prediction images of baseline model have been shown in Figure 6.

Loss Comparison

Our baseline model has established the threshold values for comparing different configurations of the model. Following our research motive, we extensively studied the performance of a segmentation model for different type of noise in the label while also investigating robustness of previously proven noise robust loss functions. To uncover the best performing loss functions, we tested all the loss functions depicted in section "Loss Function" for all the four different types of label noise while at the same

Table 1. Performance comparison of Dice, Tversky and Focal.

	Dice			Tversky			Focal		
Noise level	Accuracy (%)	IoU (%)	Loss (%)	Accuracy (%)	IoU (%)	Loss (%)	Accuracy (%)	IoU (%)	Loss (%)
10% Gaussian Noise	96.02	56.55	59.26	95.9	54.26	62.23	97.39	62.90	01.24
Mirrored Noise ($p = 0.5$)	96.94	55.61	62.81	97.22	52.10	60.43	97.21	56.10	00.59
Positive shift 20% ($p = 0.5$)	95.91	53.16	56.94	95.55	50.75	58.57	97.00	53.68	00.56
Positive shift 1% ($p = 1$)	97.25	61.52	56.37	97.25	58.76	55.01	96.98	58.78	00.62
Negative shift-20% ($p = 0.5$)	97.10	60.87	48.59	94.56	46.99	63.92	96.97	49.94	00.32
Negative shift-1% ($p = 1$)	97.20	60.42	48.08	96.52	58.09	55.32	65.61	57.06	01.35
Rotation 15 ($p = 0.5$)	96.85	60.21	51.22	96.57	53.97	58.13	97.04	57.07	00.66
Rotation 5 ($p = 1$)	96.44	54.09	56.59	96.22	45.35	66.38	96.69	52.15	00.78

time, increasing the level of artificially injected noise (gaussian, mirrored, rotation, translation) presence. To find out the best loss function among the loss functions we discussed earlier, we have gone through some systematic approaches. We have tested all the loss functions for a certain level, a level defined when the IoU score by Dice Loss goes below 50%. Afterward we tested Focal and Tversky loss for those certain levels of label noise. In *Table 1, we have showed the performance of three loss functions.*

Table 1 mentioned, the probability p indicating the percentage of data from the whole dataset has been modified by label noise (*$p = 1$ means 100% of the dataset consists of noisy label and $p = 0.5$ means 50% data consists with label noise*). A point to be noted that, Tversky loss could not provide the highest score in any single level of label noise category. As Tversky loss, could not perform better for than the other loss function, for further exploration we have omitted this loss function and try to compare with Focal and Dice Loss.

Performance under Gaussian Label Noise

First, we have applied 1%, 5%, 10%, 15%, 20%, 25% Gaussian noise to the dataset and tested them with both Dice and Focal loss. Table 2 depicted that Focal Loss performs much better than Dice Loss. In the case of 25% Gaussian noise on every label tile in the training dataset, during the testing phase Focal Loss gives 64.53% IoU where Dice Loss was giving only 9.93%. As we consider the IoU score of 50% as a threshold for an acceptable performance, we have explored over some more Gaussian noise in our Dataset. We gradually applied 30%, 40%, 50%, and 60% Gaussian noise on every image tile of the Dataset to observe declining points for noise labels. As Dice loss showed inferior performance than the Focal Loss. We have gone through a depth analysis with focal loss for more level of Gaussian noise.

Table 2. Performance comparison of Dice and Focal for Gaussian Noise.

	Dice			Focal		
Noise	Accuracy (%)	IoU (%)	F1-Score (%)	Accuracy (%)	IoU (%)	F1-Score (%)
0.01	97.57	67.51	80.60	97.19	59.66	74.73
0.05	97.64	67.30	80.46	97.75	66.58	79.94
0.10	95.95	56.55	72.24	97.50	61.33	76.03
0.15	83.07	24.04	38.76	97.48	64.58	78.48
0.20	37.16	7.96	14.76	97.72	66.02	80.38
0.25	51.64	9.93	18.07	97.63	64.53	78.44

Table 3. Performance of model trained on labels with added Gaussian noise.

Noise	Accuracy (%)	IoU (%)	F1-score (%)	Precision (%)	Recall (%)
0.10	97.50	61.33	72.10	80.40	72.10
0.20	97.79	66.02	79.10	80.98	78.13
0.30	97.16	56.83	72.47	77.38	68.15
0.40	95.99	50.33	66.96	61.03	74.17
0.50	49.38	05.48	10.39	05.75	53.61
0.60	04.39	01.48	03.56	01.88	32.23

We have injected Gaussian label noise to 10%, 20%, 30%, 40%, 50%, and 60% of pixels in each tile image. Given the noise rate as a percentage, which specifies the amount of pixels to be noisy; pixels have been chosen complete randomly and have had their label values flipped. Table 3 shows the performance of the models trained on datasets with added Gaussian noise in labels.

Increasing the noise rate from 40% to 50% causes a massive collapse of primary performance evaluated by IoU, as it drops from 50.33% to 05.48%. Considering a modest lower threshold on the IoU of 50%, results in Table 3 show that adding more than 40% Gaussian noise to the training labels can lead to inferior prediction performance. Similar trend is also observed for Accuracy, where increasing the noise rate from 40% to 50% causes a drop in accuracy of 48.55%, and likewise, F1-score score drops by 84.48%.

We observe that low noise rates yield very high recall and precision. However, as the noise rate increases, precision sharply declines, whereas recall steadily increases. This is because, our dataset is imbalanced, higher noise rates increase the proportion of water body pixels (albeit most are observed noisy pixels). The number of predicted FP pixels increases, reducing the precision as FP is in the denominator for the equation of precision, and the number of FN pixels decreases, increasing the precision as FN is in the denominator for the equation of recall. We observe that, above the noise rate of 40% Gaussian noise, the model predictions have no viable utility. Qualitative results in the form of predicted labels in Figure 7 also confirms this, as the quality of predictions is significantly reduced by training on data with more than 40% Gaussian noise added to the labels.

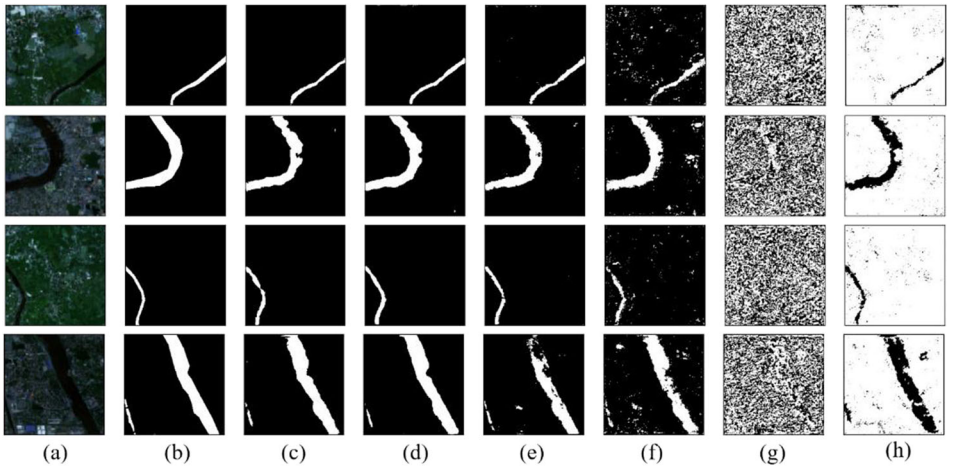


Figure 7. Predictions from model trained on labels injected with Gaussian noise (a) Image, (b) Ground truth, (c) predictions from model trained on 10% Gaussian Noise, (d) predictions from model trained on 20% Gaussian Noise, (e) predictions from model trained on 30% Gaussian Noise, (f) predictions from model trained on 40% Gaussian Noise (g) predictions from model trained on 50% Gaussian Noise and (h) predictions from model trained on 60% Gaussian Noise in the label.

Table 4. Performance comparison of dice and focal on negative shift.

		Dice			Focal		
Negative shift	Probability (noise rate)	Accuracy (%)	IoU (%)	F1-score (%)	Accuracy (%)	IoU (%)	F1-score (%)
−0.01	0.5	97.44	65.61	79.23	97.38	64.02	**78.06**
−0.05		94.11	45.71	62.74	96.53	54.21	70.30
−0.10		94.64	47.10	64.03	96.73	50.00	66.72
−0.15		95.54	52.72	69.04	96.44	43.91	61.03
−0.2		97.10	60.87	75.67	96.78	52.25	68.64
−0.01	1	97.20	60.42	75.32	97.02	59.66	74.73
−0.05		94.20	28.61	44.49	93.49	21.23	35.02
−0.10		92.29	12.71	22.56	92.63	11.61	20.81
−0.15		92.26	09.85	17.94	94.38	09.89	18.00
−0.2		92.53	05.68	10.75	94.24	05.88	11.11

Performance under Translation Induced Label Noise

We have shifted the labels by 1%, 5%, 15%, and 20% pixels with respect to the image dimensions, both positively and negatively. As an instance of translation induced noise, we first check the performance of Focal and Dice loss with negative shift. Form Table 4, we observed IoU score was fluctuating at different level of translation noise. Where Dice loss giving 65.61%, 45.71%, and 47.10% IoU score for negative shift of 1%, 5%, and 10% respectively with probability 50%, focal loss consistently performed better by giving more 50% IoU for those level. On the other hand, Dice loss performed better for negatively 15% shift with 50% probability and all the cases of negative shift with 100% probability. Therefore, we have looked to the accuracy and F1-score. For those level of noise label where Focal

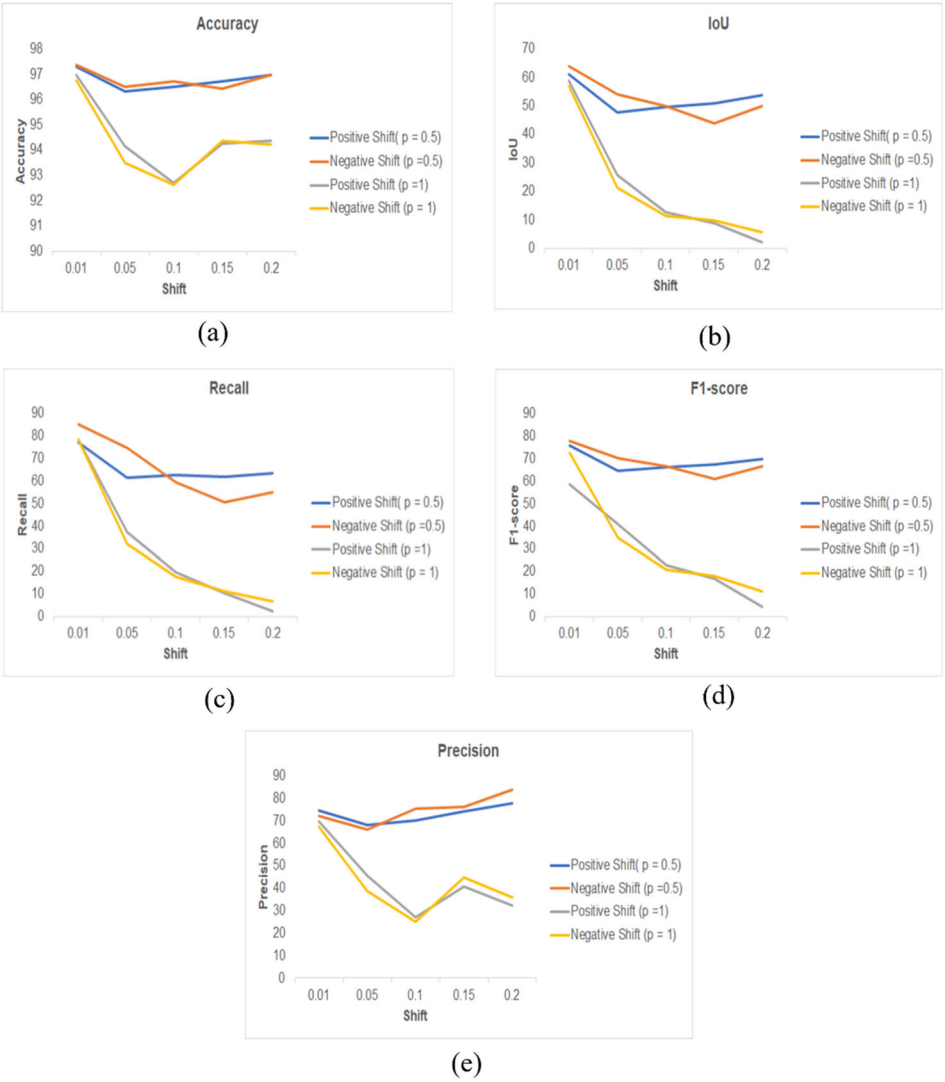


Figure 8. Performance metrics visualized as functions of translation induced label noise amount (a) accuracy, (b) IoU score, (c) recall, (d) F1-score, and (e) precision.

loss have given inferior performance to the Dice loss, the accuracy and F1-score showed better result on those points. With the increase of label noise, the amount of False negative value for pixels also increases which indicates the correct prediction capability of the loss function.

Since Focal loss performed better at less noisy level and the last above the threshold value of IoU score is at the same level of noisy label by shifting, we have decided to perform further analysis for the positive and negative shift with focal noise.

We find that the highest accuracy of 97.31% is achieved by the model with focal loss trained on data with 50% labels positively shifted by 1% (Figure 8(a)).

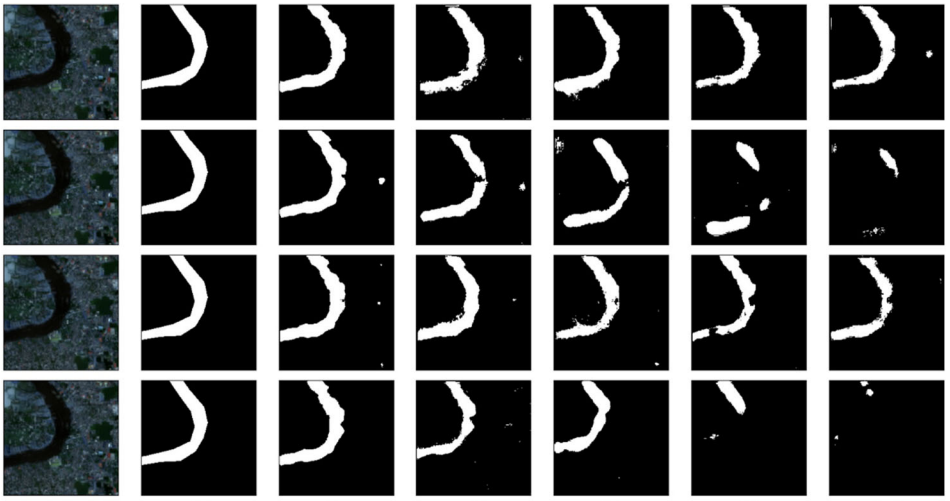


Figure 9. Predictions from models trained on data with translation induced noise (a) Image (b) Ground truth label (c) Prediction from model trained on 1% shifted label tiles (d) Prediction from model trained on 5% shifted label tiles (e) Prediction from model trained on 10% shifted label tiles (f) Prediction from model trained on 15% shifted label tiles (g) Prediction from model trained on 20% shifted labels (first row: $p = 0.5$ and positive shift, second row: $p = 1.0$ and positive shift, third row: $p = 0.5$ and negative shift, fourth row: $p = 1.0$ and negative shift).

When 50% label tiles in training data are shifted, the IoU also gradually decreases for 1% to 10% shift (Figure 8(b)). The accuracy reduces when $p = 1.0$ for positive label shift, i.e., when all label tiles in the training set are translated by 1% to 10%. The same case is observed for IoU and $p = 1.0$, as IoU drastically falls as p increases. For $p = 0.5$, when shift amount is increased from 1% to 15%, the Recall decreases (Figure 8(c)).

Recall values fall more strikingly for $p = 1.0$. Comparing Figure 8(c,e), the result of Precision has increased when the Recall value decreased and has decreased when the Recall value increased. From Figure 8(d), F1-score has showed that result reduces as we increase the noise in the training dataset by shifting the label. When training on models with negative shift and $p = 0.5$, the highest IoU is 64.02%, and accuracy is 97.38%; the value of IoU decreases with the increase of shift range. A significant change has been also observed in F1-score and Precision when pixels are negatively shifted by 1% for $p = 1.0$.

In Figure 9, we have shown the prediction results under translation induced label noise in training data for different noise rates, shift types, and shift amounts. We can see the model cannot predict the labels properly with the increase of shifting amount. When we have shifted all the labels of the dataset ($p = 1.0$), the model is trained with completely false features (with respect to the newly noised observed labels). In Figure 9, the shape of the prediction label in (c) to (g) in the second row, which is the

Table 5. Performance comparison of Dice and Focal under rotation induced label noise.

		Dice			Focal		
Rotate	Probability (noise rate)	Accuracy (%)	IoU (%)	F1-score (%)	Accuracy (%)	IoU (%)	F1-score (%)
5	0.5	97.16	57.63	73.12	97.24	60.58	75.45
10		96.32	55.07	71.02	97.10	59.63	71.72
15		96.85	60.21	75.17	96.04	57.07	72.67
5	1	96.44	54.09	70.21	96.69	52.15	68.55
10		96.14	41.93	59.09	95.92	37.75	54.79
15		95.38	39.07	56.18	95.12	36.38	53.35

Table 6. Performance metrics of models trained on data with rotation induced label noise.

Rotate	Probability (noise rate)	Accuracy (%)	**IoU (%)**	F1-score (%)	Precision (%)	Recall (%)
5	0.5	97.24	60.58	75.45	73.57	77.43
10		97.10	59.63	71.72	71.72	77.95
15		96.04	57.07	72.67	73.41	71.94
5	1	96.69	52.15	68.55	71.54	65.80
10		95.92	37.75	54.79	69.84	45.08
15		95.12	36.38	53.35	56.09	50.87

predicted labels of the 100% shifted dataset, which may appear appropriate at first glance, but the actual labels are not in that place.

Performance under Rotation Induced Label Noise

In this section, we discuss about the performance of models trained on data with rotation induced words. We have used noise rates of 0.5 and 1.0, and rotation amounts of 5, 10 and 15 degrees. We have tested all type of rotation induced model with Dice and Focal Loss separately and evaluate them on first based one the Accuracy, IoU and F1-score. IoU score is prior among other metrics for the evaluation of ta segmentation model and the threshold value of IoU for good prediction is 50%.

In Table 5, the comparative performance of both losses has been showed. In some noisy labels like 15-degree rotation with a probability 50% ($p = 0.5$), Focal loss performed slightly poor than the dice loss but for 10-degree and 5 degree rotation Focal loss overcome the Dice loss. But with 100% probability ($p = 1$) Dice again performed better but the difference between performance is not that much significant. As both of the loss functions performed their last above threshold score for IoU at the same level of noise which is 15-degree rotation with probability 50% and 1-degree rotation with 100% probability. So, further depth analysis can be done with any one of them. Since we have done Gaussian noise analysis with the focal noise we have also continued with Focal loss for this rotational noise.

The highest accuracy of 97.24% is achieved for $p = 0.5$ and rotation amount of 5 degrees (Table 6). For $p = 1.0$, the highest accuracy achieved is 96.69% and the highest IoU achieved is 52.15% (for a model trained on data with all label tiles rotated by 5 degrees). Both accuracy and IoU

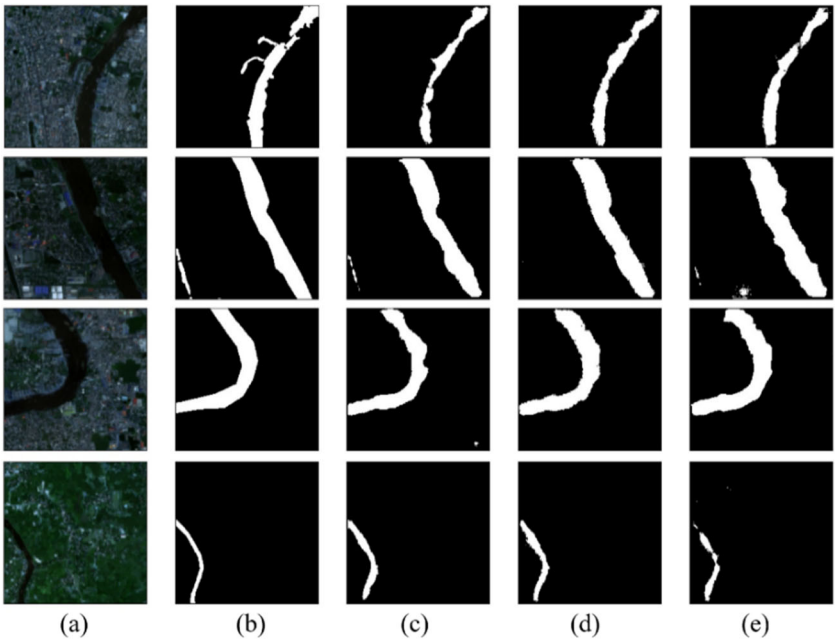


Figure 10. Predictions from models trained on data with rotation induced noise and noise rate $p = 0.5$ (a) Image (b) Ground truth label (c) Prediction from model trained on noisy labels rotated by 5 degrees (d) Prediction from model trained on noisy labels rotated by 10 degrees (e) Prediction from model trained on noisy labels rotated by 15 degrees.

reduces by 0.79%, and IoU 27.61% respectively, due to the 10-degree rotation of all the training labels. With the increase of rotational degrees, the training dataset's noise has increased for which the model could not predict accurately.

In Figure 11, we have shown the prediction after rotating all the labels 5 degrees, 10 degrees, and 15 degrees, and in Figure 10, we have shown models' predictions after rotating 50% of the labels.

Performance under Mirroring Induced Noise

We have observed the effect of different loss functions on our model for mirrored labels noisy data. In Table 7, the performance of Dice and Focal loss under mirrored label noise dataset is given. Though Dice performs better at less noisy label data, when the noise increases Focal loss slightly performs better than the Dice loss. In this case, the last noisy point where value of IoU is above the threshold is still remaining at 0.5% mirrored label noise over total train dataset for both of the noise. Here, we also have the option to explore more evaluation metrics with both of the noise but like rotation here we have explored with the Focal loss function.

In this section, we present and discuss the effects of mirrored labels in training data on model performance (Table 8). As expected, for the best

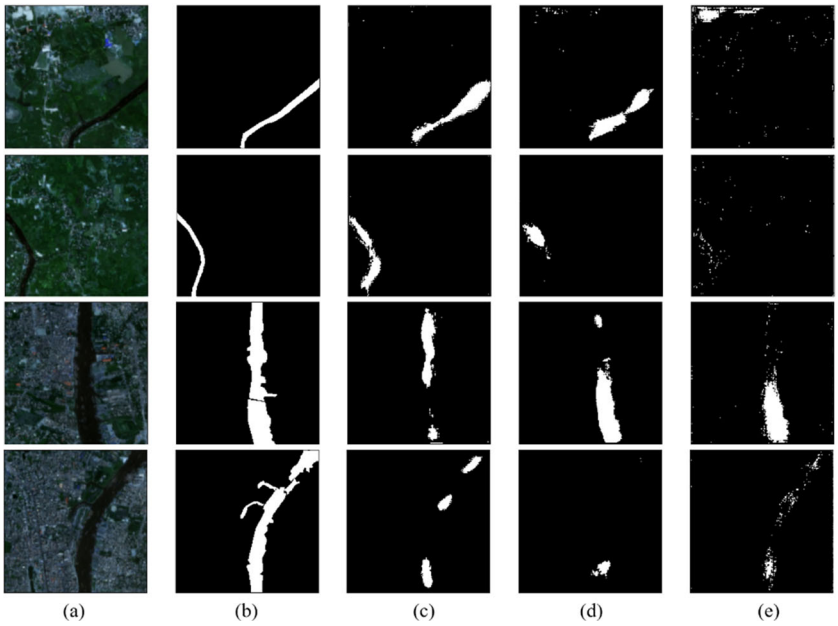


Figure 11. Predictions from models trained on data with rotation induced noise and noise rate $p = 1.0$ ((a) Image (b) Ground truth label (c) Prediction from model trained on noisy labels rotated by 5 degrees (d) Prediction from model trained on noisy labels rotated by 10 degrees (e) Prediction from model trained on noisy labels rotated by 15 degrees.

Table 7. Performance comparison of Dice and Focal under mirroring induced noise.

	Dice			Focal		
Probability (noise rate)	Accuracy (%)	IoU (%)	F1-score (%)	Accuracy (%)	IoU (%)	F1-score (%)
0.01	97.74	67.79	80.80	97.35	60.18	75.14
0.05	96.94	55.61	71.48	97.21	56.10	71.88
0.10	94.60	02.52	04.91	94.09	02.90	05.65

Table 8. Performance metrics of model trained on data with mirrored labels.

Probability (noise rate)	Accuracy (%)	IoU (%)	F1-score (%)	Precision (%)	Recall (%)
0.25	97.35	60.18	75.14	77.35	73.05
0.5	97.21	56.10	71.88	80.57	64.88
1	94.09	02.90	05.65	22.56	03.23

performance is for $p = 0.25$, where only 25% of randomly selected label tiles are mirrored; the model achieves IoU of 60.18% and accuracy of 97.35%. The highest Precision of 77.35% is also for the training data with $p = 0.25$. However, when $p = 1.00$, i.e. when all label tiles in training data are mirrored and noise, the accuracy drops to 3.31%. Again, this behavior is expected as training on data with only noisy labels should not even yield a modestly well-performing model. Comparing performance for $p = 0.5$ and $p = 1.0$, the latter induces IoU drop of 94.83% and Recall drop of 95.02%, compared to the former. Figure 12 shows the predictions made by models on some test set images.

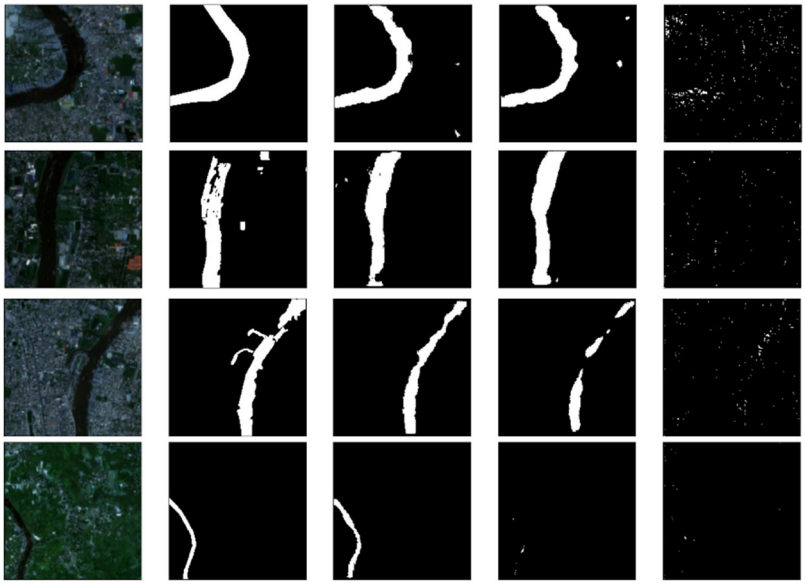


Figure 12. Predictions made by models trained on data with mirrored label tiles (a) Image (b) Ground truth label (c) Predictions from model trained on data with mirrored labels ($p = 0.25$), (d) Predictions from model trained on data with mirrored labels ($p = 0.5$) (e) Predictions from model trained on data with mirrored labels ($p = 1.0$).

These qualitative results show that model predictions remain coherent for $p = 0.25$ and $p = 0.5$, but for $p = 1.0$ the qualitative results were very poor.

Conclusions

In this work, we have evaluated the effects of various types and settings of label noise on performance of remote sensing and deep learning-based water body segmentation models.

We first find the baseline ideal scenario where the model is trained solely on data with clean and noise free labels. We have observed that, for Gaussian label noise, noise rates below or equal to approximately 40% with Focal loss can still be good enough for learning representations from. However, for Gaussian noise rate of 50% and more, the overall performance falls significantly. For mirroring induced label noise, $p = 0.5$ still gives satisfactory results. For rotation induced label noise, $p = 0.5$ and rotation amount of up to 15 degrees have been observed to be tolerable, whereas for $p = 1.0$, and rotation amount above 5% does not provide acceptable performance. We have also evaluated translation induced label noise.

There are several potential endpoints open for further investigation. Firstly, the study area size can be increased by taking other cities facing similar issues, or perhaps can be carried out on national scales due to high

availability and coverage of Sentinel-2 imagery, which will focus more extensively on water bodies across our country. Other network architectures such as DeepLabV3, Gated SCNN can also be explored as the segmentation model in lieu of the U-Net or can be directly compared to each other. Other settings might involve improving the dataset by having multiple labels for the same region, labeled by different labelers with varying levels of expertise, and then use the aggregated data to make more informed and stronger inferences. Feature noise could be another area of exploration in future.

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References

Angel, S., J. Parent, D. L. Civco, and A. M. Blei. 2012. Administrative Boundaries, Dhaka, Bangladesh, 1990. [Shapefile]. *Lincoln Institute of Land Policy*. https://earthworks.stanford.edu/catalog/stanford-cc486qp3429

Afroz, R. 2016. *Impact of green roof on urban canopy layer microclimates in planned residential area of Dhaka, Bangladesh. Ass. Builiding Green Futures, Bolgona : 150* .

Carroll, M., M. Wooten, C. DiMiceli, R. Sohlberg, and M. Kelly. 2016. Quantifying surface water dynamics at 30 meter spatial resolution in the North American high northern latitudes 1991–2011. *Remote Sensing* 8 (8):622. doi:10.3390/rs8080622.

Cheikh Sidiya, A., and X. Li. 2020. Style-based unsupervised learning for real-world face image super-resolution. *Recent advances in image restoration with applications to real world problems* 71. doi:10.5772/intechopen.92320.

Chen, Y., R. Fan, X. Yang, J. Wang, and A. Latif. 2018. Extraction of urban water bodies from high-resolution remote-sensing imagery using deep learning. *Water* 10 (5):585. doi: 10.3390/w10050585.

Chu, Z., T. Tian, R. Feng, and L. Wang. 2019. Sea-land segmentation with Res-UNet and fully connected CRF, *IGARSS 2019-2019 IEEE International Geoscience and Remote Sensing Symposium,* 3840–3. doi:10.1109/igarss.2019.8900625.

Du, G., X. Cao, J. Liang, X. Chen, and Y. Zhan. 2020. Medical image segmentation based on U-Net: A review. *Journal of Imaging Science and Technology* 64 (2):20508–1–12. doi: 10.2352/J.ImagingSci.Technol.2020.64.2.020508.

Du, Y., Y. Zhang, F. Ling, Q. Wang, W. Li, and X. Li. 2016. Water bodies' mapping from Sentinel-2 imagery with modified normalized difference water index at 10-m spatial resolution produced by sharpening the swir band. *Remote Sensing* 8 (4):354. doi:10.3390/rs8040354.

Faisal, I. M., and S. Parveen. 2004. Food security in the face of climate change, population growth, and resource constraints: Implications for Bangladesh. *Environmental Management* 34 (4):487–98. doi:10.1007/s00267-003-3066-7.

Falk, T., D. Mai, R. Bensch, Ö. Çiçek, A. Abdulkadir, Y. Marrakchi, A. Böhm, J. Deubner, Z. Jäckel, K. Seiwald, et al. 2019. U-Net: Deep learning for cell counting, detection, and morphometry. *Nature Methods* 16 (1):67–70. doi:10.1038/s41592-018-0261-2.

Feng, W., H. Sui, W. Huang, C. Xu, and K. An. 2019. Water body extraction from very high-resolution remote sensing imagery using deep U-Net and a superpixel-based conditional random field model. *IEEE Geoscience and Remote Sensing Letters* 16 (4):618–22. doi:10.1109/LGRS.2018.2879492.

Frank, J., U. Rebbapragada, J. Bialas, T. Oommen, and T. Havens. 2017. Effect of label noise on the machine-learned classification of earthquake damage. *Remote Sensing* 9 (8): 803. doi:10.3390/rs9080803.

Frénay, B., and M. Verleysen. 2014. Classification in the presence of label noise: A survey. *IEEE Transactions on Neural Networks and Learning Systems* 25 (5):845–69. doi:10.1109/TNNLS.2013.2292894.

Gaál, G., B. Maga, and A. Lukács. 2020. Attention U-Net based adversarial architectures for chest X-ray lung segmentation. *arXiv preprint arXiv:2003.103042692*

Ghosh, A., H. Kumar, and P. S. Sastry. 2017. Robust loss functions under label noise for deep neural networks. *Proceedings of the AAAI Conference on Artificial Intelligence* 31: 1919–1925.

Goldblatt, R., W. You, G. Hanson, and A. Khandelwal. 2016. Detecting the boundaries of urban areas in India: A dataset for pixel-based image classification in Google Earth Engine. *Remote Sensing* 8 (8):634. doi:10.3390/rs8080634.

Gorelick, N., M. Hancher, M. Dixon, S. Ilyushchenko, D. Thau, and R. Moore. 2017. Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment* 202:18–27. doi:10.1016/j.rse.2017.06.031.

Huang, Q., J. Sun, H. Ding, X. Wang, and G. Wang. 2018. Robust liver vessel extraction using 3D U-Net with variant dice loss function. *Computers in Biology and Medicine* 101: 153–62. doi:10.1016/j.compbiomed.2018.08.018.

Isikdogan, F., A. C. Bovik, and P. Passalacqua. 2017. Surface water mapping by deep learning. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 10 (11):4909–18. doi:10.1109/JSTARS.2017.2735443.

Jadon, S., G. Wang, X. Liu, C. Li, Z. Xu, J. Ruan, H. Zhu, T. Meng, K. Li, N. Huang, et al. 2020. A survey of loss functions for semantic segmentation. *IEEE Access 8*(8), 1919–1925. doi:10.1109/CIBCB48159.2020.9277638.

Jansson, A., E. Humphrey, N. Montecchio, R. Bittner, A. Kumar, and T. Weyde. 2017. *Singing voice separation with deep U-Net convolutional networks. 18th International Society for Music Information Retrieval Conference*, 23-27 Oct 2017, Suzhou, China.

Jawak, S. D., K. Kulkarni, and A. J. Luis. 2015. A review on extraction of lakes from remotely sensed optical satellite data with a special focus on cryospheric lakes. *Advances in Remote Sensing* 04 (03):196–213. doi:10.4236/ars.2015.43016.

Li, J., X. Huang, and X. Chang. 2020. A label-noise robust active learning sample collection method for multi-temporal urban land-cover classification and change analysis. *ISPRS Journal of Photogrammetry and Remote Sensing* 163:1–17. doi:10.1016/j.isprsjprs.2020.02.022.

Lin, T. Y., P. Goyal, R. Girshick, K. He, and P. Dollar. 2020. Focal loss for dense object detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 42 (2):318–27. doi:10.1109/TPAMI.2018.2858826.

Ma, Y. D., Q. Liu, and Z. B. Qian. 2004. Automated image segmentation using improved PCNN model based on cross-entropy. *Proceedings of 2004 International Symposium on Intelligent Multimedia, Video and Speech Processing,* 2004:743–6. doi:10.1109/isimp.2004.1434171.

Maksud Kamal, A. S. M., and S. Midorikawa. 2004. GIS-based geomorphological mapping using remote sensing data and supplementary geoinformation. A case study of the Dhaka city area. *International Journal of Applied Earth Observation and Geoinformation* 6 (2):111–25. doi:10.1016/j.jag.2004.08.003.

Ngugi, L. C., M. Abdelwahab, and M. Abo-Zahhad. 2020. Tomato leaf segmentation algorithms for mobile phone applications using deep learning. *Computers and Electronics in Agriculture* 178:105788. doi:10.1016/j.compag.2020.105788.

O'Shea, K., and R. Nash. 2015. *An introduction to convolutional neural networks*. arXiv: 1511.08458

Rahman, S., and F. Hossain. 2008. Spatial assessment of water quality in Peripheral Rivers of Dhaka City for optimal relocation of water intake point. *Water Resources Management* 22 (3):377–91. doi:10.1007/s11269-007-9167-y.

Ronneberger, O., P. Fischer, and T. Brox. 2015. U-net: Convolutional networks for biomedical image segmentation. *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)* 9351:234–41. doi:10.1007/978-3-319-24574-4_28.

Salehi, S. S., A. G. Deniz Erdogmus, and A. Gholipour. 2017. Tversky loss function for image segmentation using 3D fully convolutional deep networks. 10541:106–13. doi:10.1007/978-3-319-67389-9.

Siddique, N., P. Sidike, C. Elkin, and V. Devabhaktuni. 2020. *U-Net and its variants for medical image segmentation: theory and applications.* arXiv preprint arXiv:2011.01118.

Smith, A. G., J. Petersen, R. Selvan, and C. R. Rasmussen. 2020. Segmentation of roots in soil with U-Net. *Plant Methods* 16 (1):13–5. doi:10.1186/s13007-020-0563-0.

Wang, G., M. Wu, X. Wei, and H. Song. 2020. Water identification from high-resolution remote sensing images based on multidimensional densely connected convolutional neural networks. *Remote Sensing* 12 (5):795. doi:10.3390/rs12050795.

Wang, G., X. Liu, C. Li, Z. Xu, J. Ruan, H. Zhu, T. Meng, K. Li, N. Huang, and S. Zhang. 2020. A noise-robust framework for automatic segmentation of COVID-19 pneumonia lesions from CT images. *IEEE Transactions on Medical Imaging* 39 (8):2653–63. doi:10.1109/TMI.2020.3000314.

Wang, J., J. Lu, G. Qin, L. Shen, Y. Sun, H. Ying, Z. Zhang, and W. Hu. 2018. Technical note: A deep learning-based autosegmentation of rectal tumors in MR images. *Medical Physics* 45 (6):2560–4. doi:10.1002/mp.12918.

Wang, L., C. Wang, Z. Sun, and S. Chen. 2020. An improved dice loss for pneumothorax segmentation by mining the information of negative areas. *IEEE Access.* 8:167939–49. doi:10.1109/ACCESS.2020.3020475.

Weng, Q. 2002. Land use change analysis in the Zhujiang Delta of China using satellite remote sensing, GIS and stochastic modelling. *Journal of Environmental Management* 64 (3):273–84. doi:10.1006/jema.2001.0509.

Xu, H. 2006. Modification of normalised difference water index (NDWI) to enhance open water features in remotely sensed imagery. *International Journal of Remote Sensing* 27 (14):3025–33. doi:10.1080/01431160600589179.

Yang, X., S. Zhao, X. Qin, N. Zhao, and L. Liang. 2017. Mapping of urban surface water bodies from Sentinel-2 MSI imagery at 10 m resolution via NDWI-based image sharpening. *Remote Sensing* 9 (6):596. doi:10.3390/rs9060596.

Zeng, C., S. Bird, J. Luce, and J. Wang. 2015. A natural-rule-based-connection (NRBC) method for river network extraction from high-resolution imagery. *Remote Sensing* 7 (10):14055–78. doi:10.3390/rs71014055.

Zhang, P., Y. Ke, Z. Zhang, M. Wang, P. Li, and S. Zhang. 2018. Urban land use and land cover classification using novel deep learning models based on high spatial resolution satellite imagery. *Sensors* 18 (11):3717. doi:10.3390/s18113717.

Zhang, Z., Q. Liu, and Y. Wang. 2018. Road extraction by deep residual U-Net. *IEEE Geoscience and Remote Sensing Letters* 15 (5):749–53. doi:10.1109/LGRS.2018.2802944.

Zheng, Q., H. Delingette, N. Duchateau, and N. Ayache. 2018. 3-D consistent and robust segmentation of cardiac images by deep learning with spatial propagation. *IEEE Transactions on Medical Imaging* 37 (9):2137–48. doi:10.1109/TMI.2018.2820742.